



Department of Mathematics, Physics and Computer Science
College of Science and Mathematics

History of the Internet

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Learning Objectives

At the end of the lesson, all of you should be able to:

1. Define and describe the different types of computer networks
2. Describe the historical development of the Internet, including its origins in ARPANET
3. Differentiate between types of network such as
 - LANs (Local Area Networks),
 - CANs (Campus Area Networks)
 - WANs (Wide Area Networks),
 - MANs (Metropolitan Area Networks), and their respective use cases.
4. Define what an Intranet and Extranet are

What is internet?

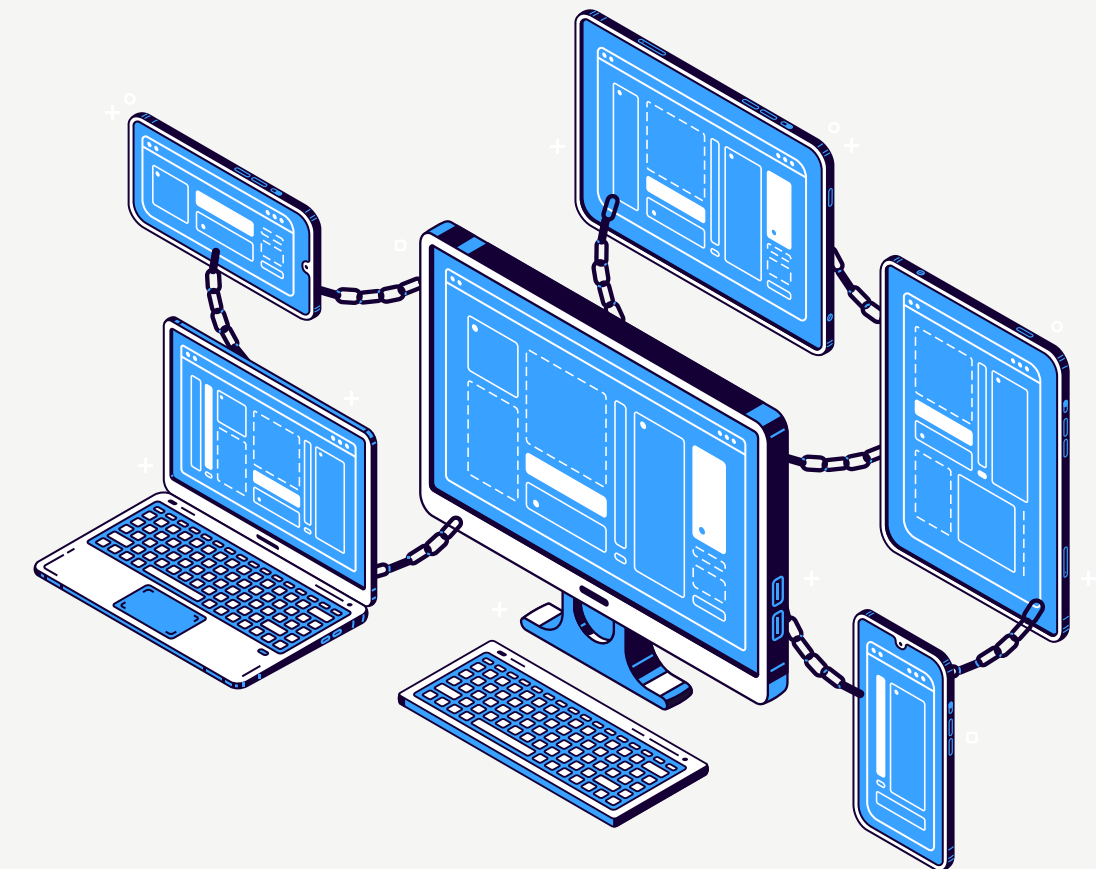
The Internet is a vast **network** that connects computers all over the world. Through the Internet, people can share information and communicate from anywhere with an Internet connection.

Key Milestones in Internet History

1969	Launch of ARPANET , the first operational packet-switching network.
1971	The first email is sent by Ray Tomlinson .
1973/1983	TCP/IP was developed and adopted.
1984	Domain Name System (DNS) was introduced.
1991	World Wide Web was created.
1993	Mosaic Web Browser was introduced.
1995	Internet commercialization began.
2000s–Present	Rise of cloud computing , social media , and streaming services .

What is Computer Network?

A computer network is defined as a **system that connects two or more computing devices for transmitting and sharing information and resources** (e.g. file, storage, printer, processing power, etc.).



Types of Computer Network

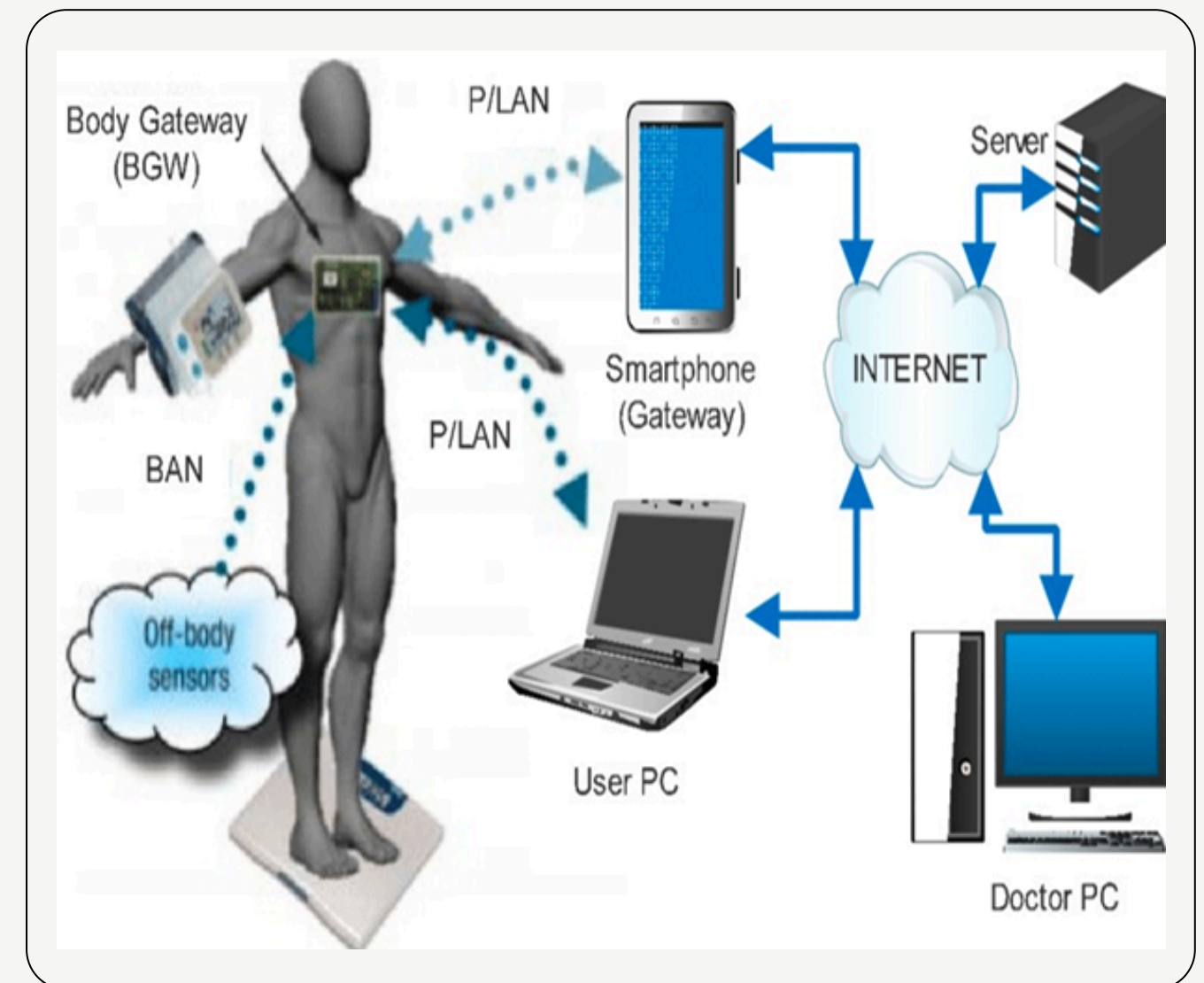
1. Personal Area Network (PAN) or Wireless PAN (WPAN)

- The smallest computer-based networks
- When your computer is connected to a device (e.g. printer) using a communication line/ link or via wireless connection.
- They can connect a wireless keyboard, mouse, or other devices to a computer.
- You may find them wirelessly linking a printer to your computer.



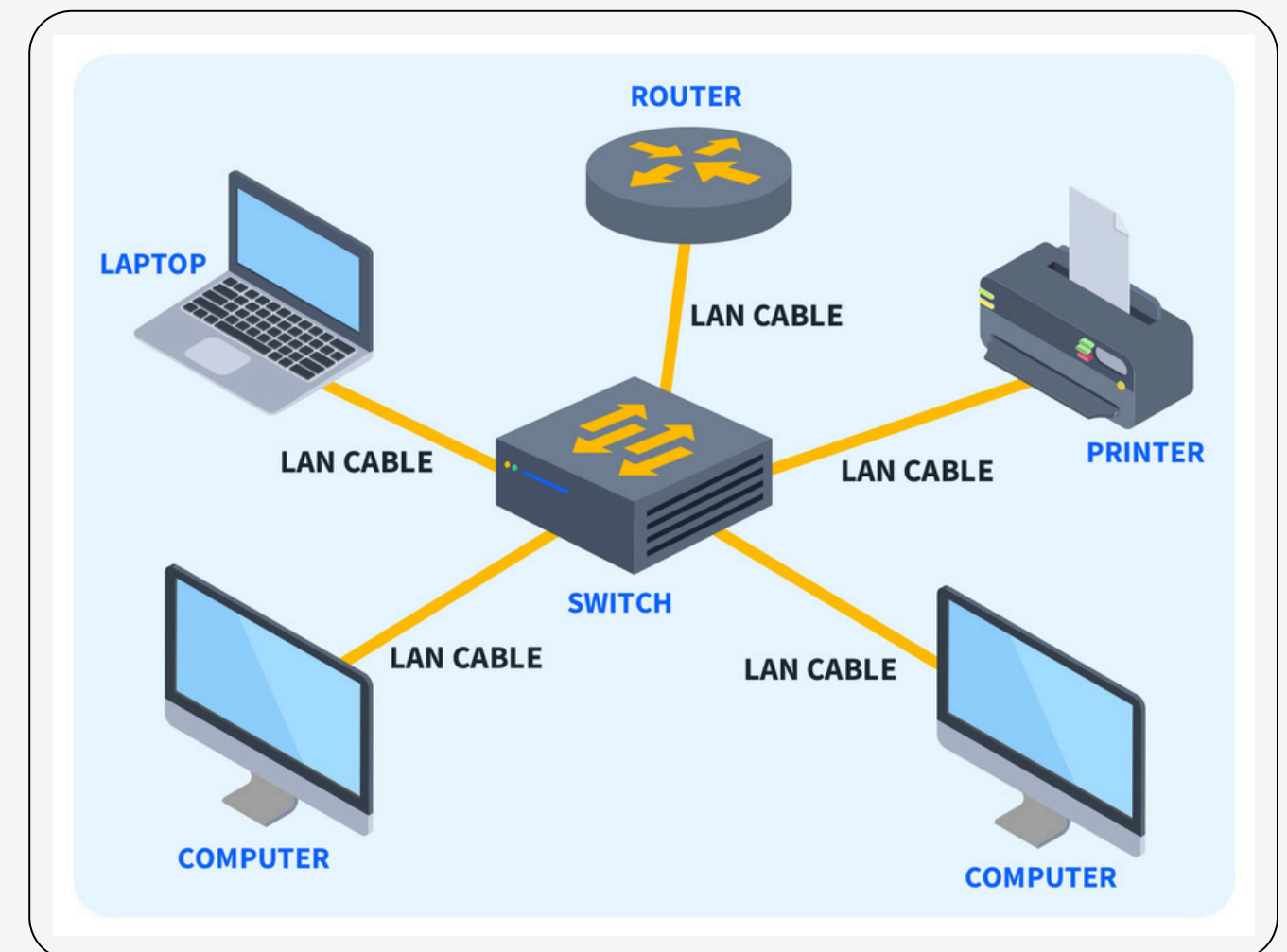
2. Body Area Network (BAN)

- A BAN is a short-range wireless network comprised of devices positioned in, on, and around the body.
- It provides data communication over short distances, limited to ranges of just a few meters.
- This network uses wearable and implanted electronic circuits.
- It implements highly useful functions and capabilities in convenient, unobtrusive configurations that operate at very low power and deliver superlative security.



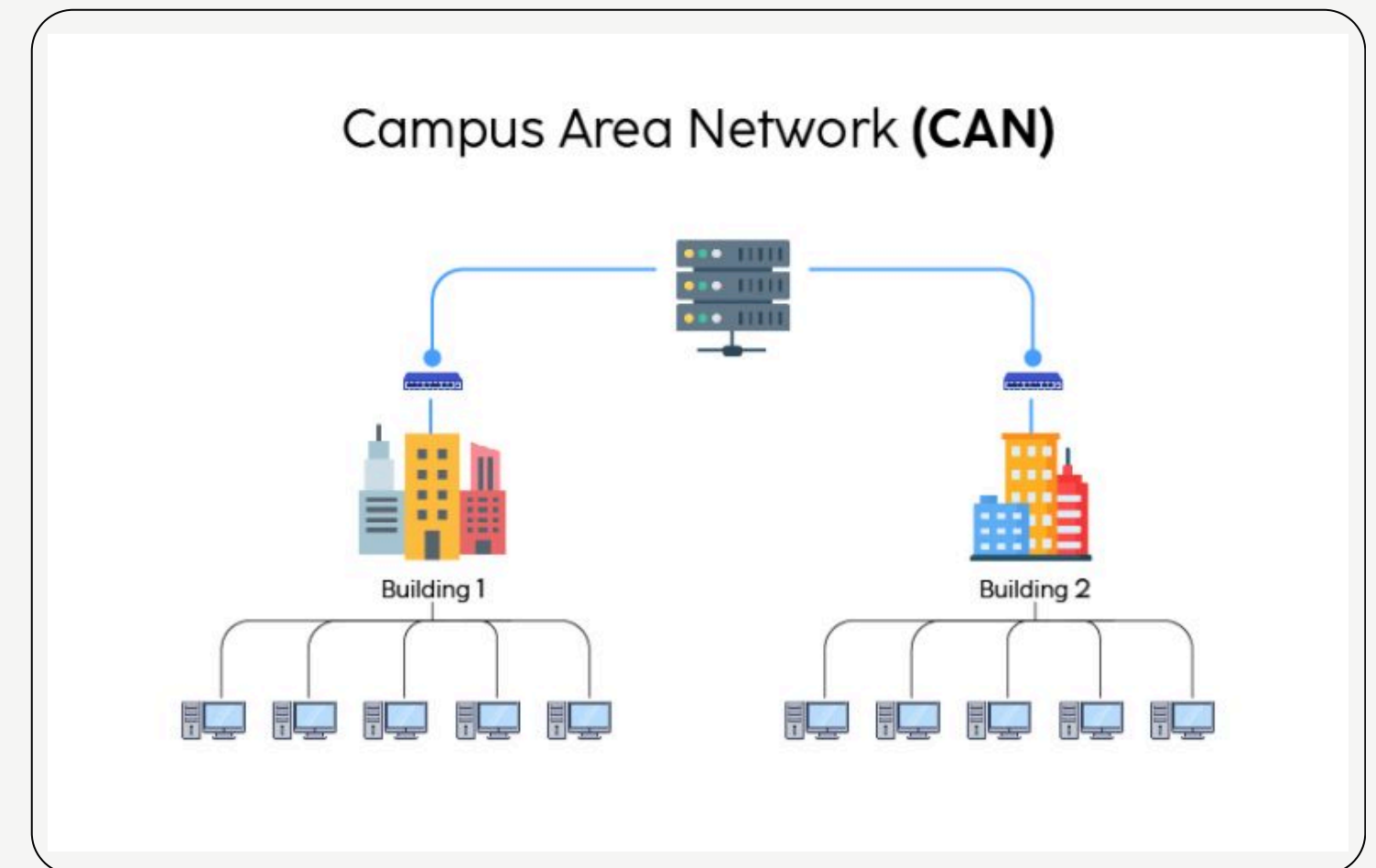
3. Local Area Network (LAN)

- is a group of computers and network devices connected together, usually within the same building (max length from the switch = 100 m).
- In UP Mindanao, the connections are either via Ethernet (10 Mbps) or Fast Ethernet (100 Mbps), Gigabit Ethernet (1000 Mbps), or via wifi.



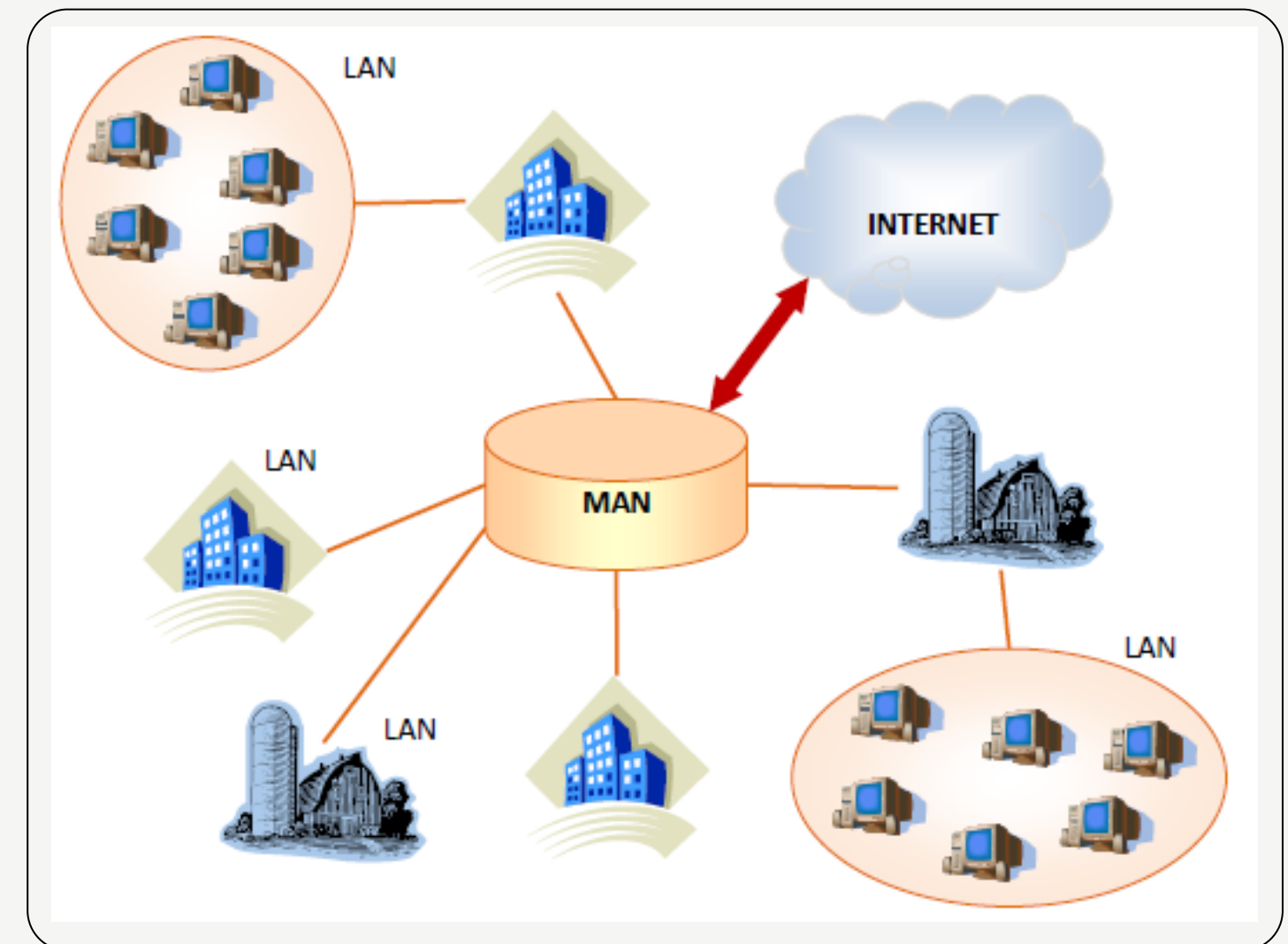
4. Campus Area Network (CAN)

- A campus network, campus area network, corporate area network or CAN is a computer network made up of an interconnection of local area networks (LANs) within a limited geographical area.
- Interconnect LANs in the same building or group of buildings



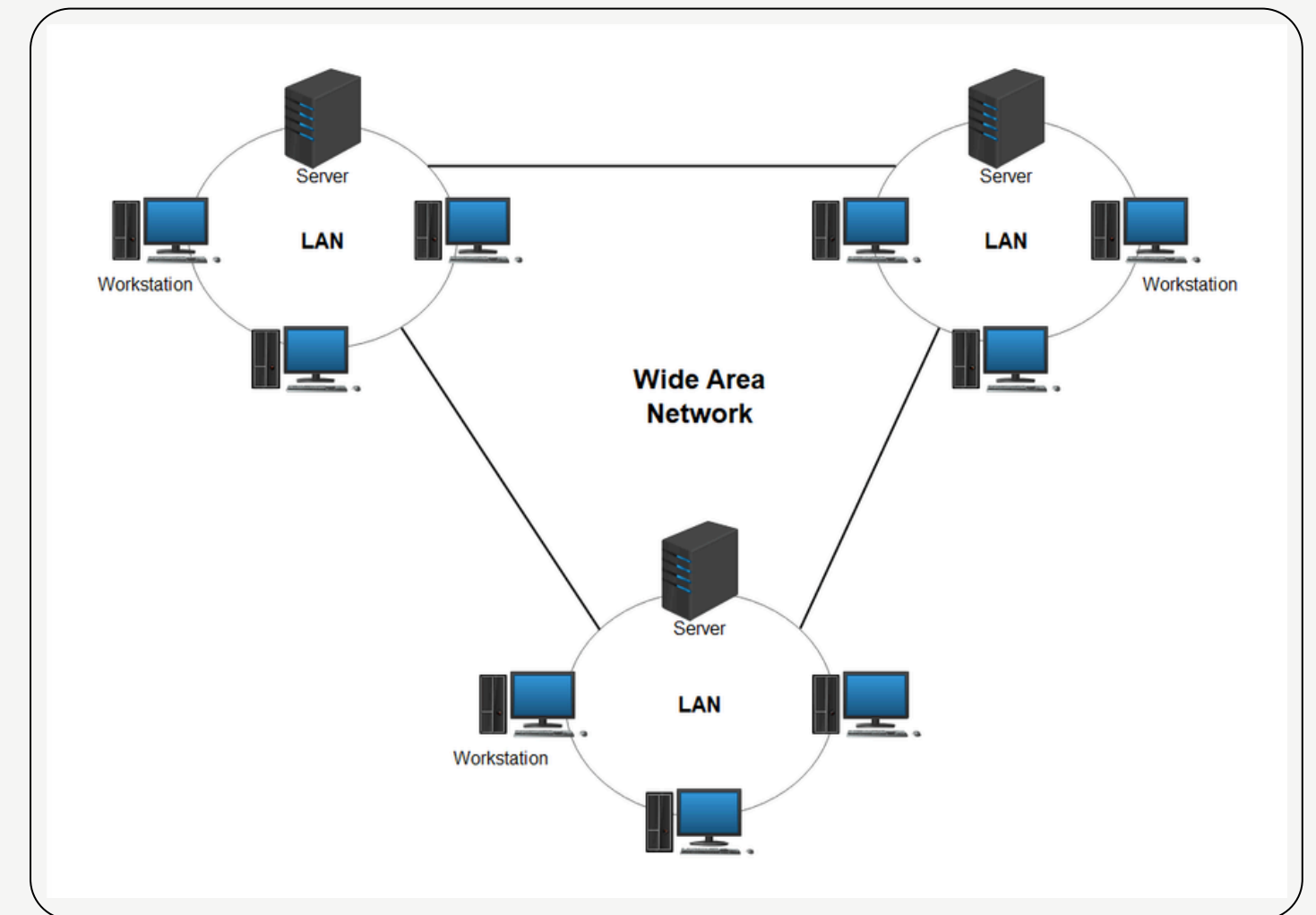
5. Metropolitan Area Network (MAN)

- It is a larger network that usually spans several buildings in the same city or town.
- It connects geographically dispersed LANs.
- Usually established using optical fiber.
- The network is established using routers and switches.
- Headquarters, R&D, manufacturing, and shipping can all be in separate locations and yet communicate as if they were in the same building.



6. Wide Area Network (WAN)

- it is a geographically distributed private telecommunications network that interconnects multiple LANs.
- In an enterprise (e.g. UP System, PLDT, BDO), a WAN may consist of connections to a company's headquarters, branch offices, co-location facilities, cloud services and other facilities.
- LANs usually connect to WAN through a router because the LAN rules for communicating (known as protocols) are different from WAN protocols.



7. The Internet

- an electronic communications network that connects computer networks and organizational computer facilities around the world (Merriam Webster Dictionary).
- The internet is a pervasive, fundamental part of daily life that continues to deliver massive economic and social benefits around the world.
- It interconnects thousands of millions of computing devices including PCs, Laptops, Workstations, Servers, Smartphones, Tablets, TVs, Webcams, Environmental devices, Automobiles, Security cameras, and many more.



Intranet

- It is a computer network that is controlled and administered by a single entity such as a company or an organization.
- In Intranet, external users are not allowed to connect.
- Usually, in this network, proper authorization is required to access any resources.
- Further, each access is monitored and logged to ensure that only authorized users get access.

Extranet

- An Extranet is an extended Intranet where external users are allowed to access a small portion of the network.
- Normally, this access is filtered and secured by several security implementations on both hardware and software levels.

Thank you!

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